A Roadmap: Functionalized Mxene for Energy Storage Applications

A Dissertation

Submitted in partial fulfillment for the award of the degree of
Master of Science (Chemistry)

Submitted by
Kumkum
Enrol no.23BSC3CH009

Under Supervision of
Dr. Nutan Sharma
Professor

Department of Chemistry
Vivekananda Global University, Jaipur

2025

VIVEKANANDA GLOBAL UNIVERSITY, JAIPUR

DISSERTATION APPROVAL CERTIFICATE

Department of Sciences, Faculty of Basic and Applied Sciences

This dissertation, titled **"A Roadmap: Functionalized Mxene for Energy Storage Applications,"** prepared and submitted by **Kumkum**, bearing University Enrolment Number **23BSC3CH009,** is duly approved for the Degree of Master of Science in Chemistry from Vivekananda Global University, Jaipur.

Supervisor
Dr. Nutan Sharma
Department of Sciences
VGU, Jaipur-303012

Prof. (Dr.) Sushila
HOD, Chemistry, VGU

Prof. (Dr.) Kailash Agarwal

Dean, FBAS

Place: Jaipur

Date:

VIVEKANANDA GLOBAL UNIVERSITY, JAIPUR

DECLARATION

I hereby declare that the project report entitled **"A Roadmap: Functionalized Mxene for Energy Storage Applications"** is my work and has been carried out under the guidance of **Dr. Nutan Sharma,** Professor, Department of Chemistry, **Vivekananda Global University, Jaipur**. This is my original work, and all steps have been taken to make this review error-free.

Name of Student: Kumkum

Program**: MSc. Chemistry**

Enrolment No. 23BSC3CH009

I hereby verify that the report submitted titled "**A Roadmap: Functionalized Mxene for Energy Storage Applications"** has undergone a plagiarism check using **Turnitin**. The result of the check indicates that the overall similarity score was less than 20%.

Dr. Nutan Sharma

professor

Department of Chemistry

Vivekananda Global University, Jaipur

ACKNOWLEDGEMENT

I acknowledge gratitude with a sense of reverence to the almighty God. I would like to express my deep regards to our esteemed **Dean, Prof. (Dr.) Kailash Agarwal** and **Dr. Sushila (Head of FBAS)** of "Vivekananda Global University, Jaipur". I would like to express my sincere thanks to my project supervisor, **Dr. Nutan Sharma (Professor)**, for giving me an opportunity to work on a valuable project and giving me her precious time. I would also like to thank other teachers, who supported me during my Master of Chemistry and offered me deep insights into study. I am also highly grateful to my parents, who encouraged me throughout my study and provided moral support.

Thank you for the support, love, encouragement, and blessings that you bestowed upon me.

Place: Jaipur
Date: 06.06.2025

Name: Kumkum
Enrolment No. 23BSC3CH009

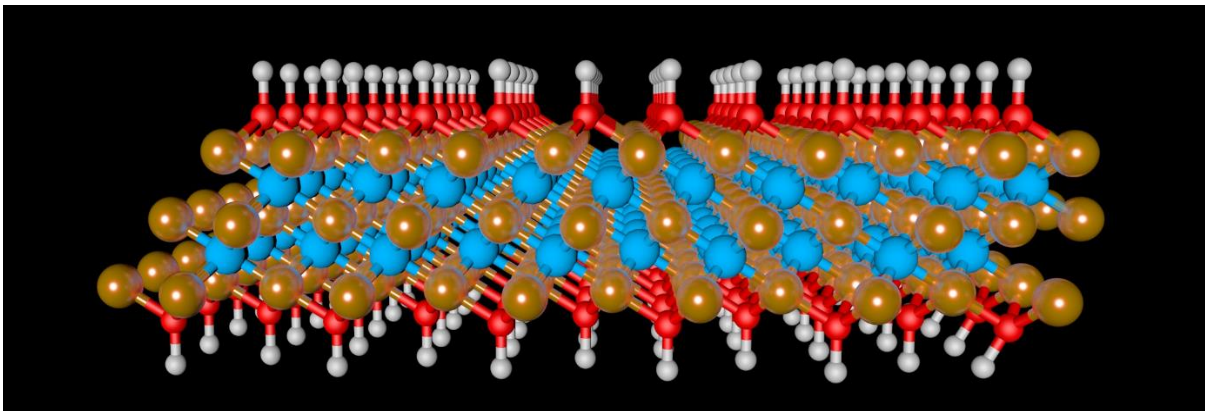
CONTENTS

ABSTRACT

MXenes, a family of two-dimensional transition metal carbides and nitrides, have emerged as highly promising materials for next-generation energy storage devices due to their remarkable electrical conductivity, tunable surface chemistry, and layered structure. The performance of MXenes can be significantly enhanced through surface functionalization, which tailors their physical and chemical interactions with various electrolytes and active materials. This paper presents a detailed overview of functionalized MXenes, exploring their synthesis, structural diversity, and role in supercapacitors, batteries, and hybrid energy storage systems. Drawing insights from recent developments and key literature, including the works of Rizwan and Gogotsi, this study outlines current challenges and the future scope of functionalized MXenes in sustainable energy applications.

CHAPTER 1

INTRODUCTION



INTRODUCTION

1.1 2D Materials

When studying the behavior of materials, their composition is often the first factor considered. However, the size and structure of a material can also play a major role in determining its properties, especially when dimensions are reduced to the nanoscale [1]. Materials at this scale often display behavior that differs significantly from their bulk counterparts, including changes in conductivity, strength, and chemical reactivity [2].

Two-dimensional (2D) materials are a unique class of nanomaterials that consist of atomically thin layers. Among the most well-known examples is graphene, a single layer of carbon atoms arranged in a hexagonal lattice, first isolated in 2004. Since then, the family of 2D materials has expanded to include a variety of structures with different compositions and functionalities [3].

What makes 2D materials so appealing is their high surface area and the ease with which their properties can be engineered. These materials are often derived from bulk precursors that can be exfoliated into individual sheets. Their thin nature allows them to interact efficiently with other materials or external stimuli, making them ideal candidates for applications in electronics, catalysis, and energy storage. MXenes, as a more recent addition to this category, bring additional versatility due to their metallic conductivity and the presence of functional surface terminations [3].

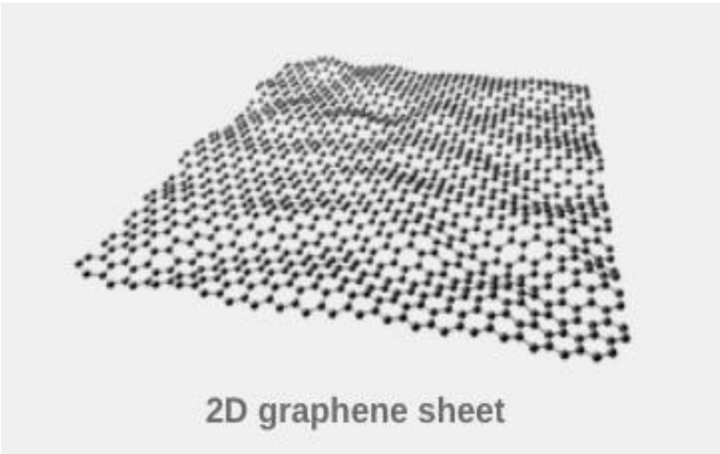


Figure 1: *2D Graphene Sheet* by Wang, X., Tang, F., Cao, Q., Qi, X., Pearson, M., Li, M., Pan, H., Zhang, Z., & Lin, Z. (2020), illustration. Accessed June 4, 2025, from https://doi.org/10.3390/nano10050838 [62]

1.2 Discovery and History of Mxene

MXenes belong to a relatively young but rapidly growing family of 2D materials. They were first synthesized in 2011 by selectively etching out the A-layer from MAX phases —-layered ceramics with the general formula Mn+1AXn. In these structures, “M” is an early transition metal, “A” is usually a group 1 of 3 elements Al or Si), and “X” is carbon and/or nitrogen [4].

The key to producing MXenes lies in removing the A-element using chemical methods, typically with hydrofluoric acid or fluoride-containing salts. What's left is a 2D sheet of transition metal carbides or nitrides that exhibits excellent conductivity, a large surface area, and reactive surfaces that can be further functionalized [5].

Initially, MXenes were studied primarily for their electronic and mechanical properties. However, their hydrophilic nature and surface terminations (–OH, –O, and –F) made them attractive for electrochemical applications. Over the past decade, their use has expanded into supercapacitors, batteries, electromagnetic shielding, sensors, and even biomedical applications. This growing versatility has solidified MXenes as one of the most promising classes of functional nanomaterials [5].

1.3 Structure of Mxene

The atomic structure of MXenes is directly inherited from their MAX phase precursors. After the selective removal of the A-layer, what remains is a layered material where X atoms (carbon or nitrogen) are sandwiched between layers of metal atoms (M). This results in a sheet-like structure similar in appearance to graphene, but with more complexity due to the presence of surface terminations.

In a typical MXene unit, each M-X bond is strong and stable, and the entire sheet adopts a hexagonal lattice. These layers tend to restack due to van der Waals forces, but this can be controlled through intercalation or surface modifications. The surface terminations also play a crucial role, not just in defining chemical reactivity, but also in influencing conductivity, hydrophilicity, and ion accessibility [6].

As researchers continue to explore new MXene compositions, structural variations have been observed, including double transition metal MXenes and ordered/disordered surface patterns. These structural traits make MXenes highly tunable and suitable for energy storage, where both surface chemistry and ion mobility are key factors [7].

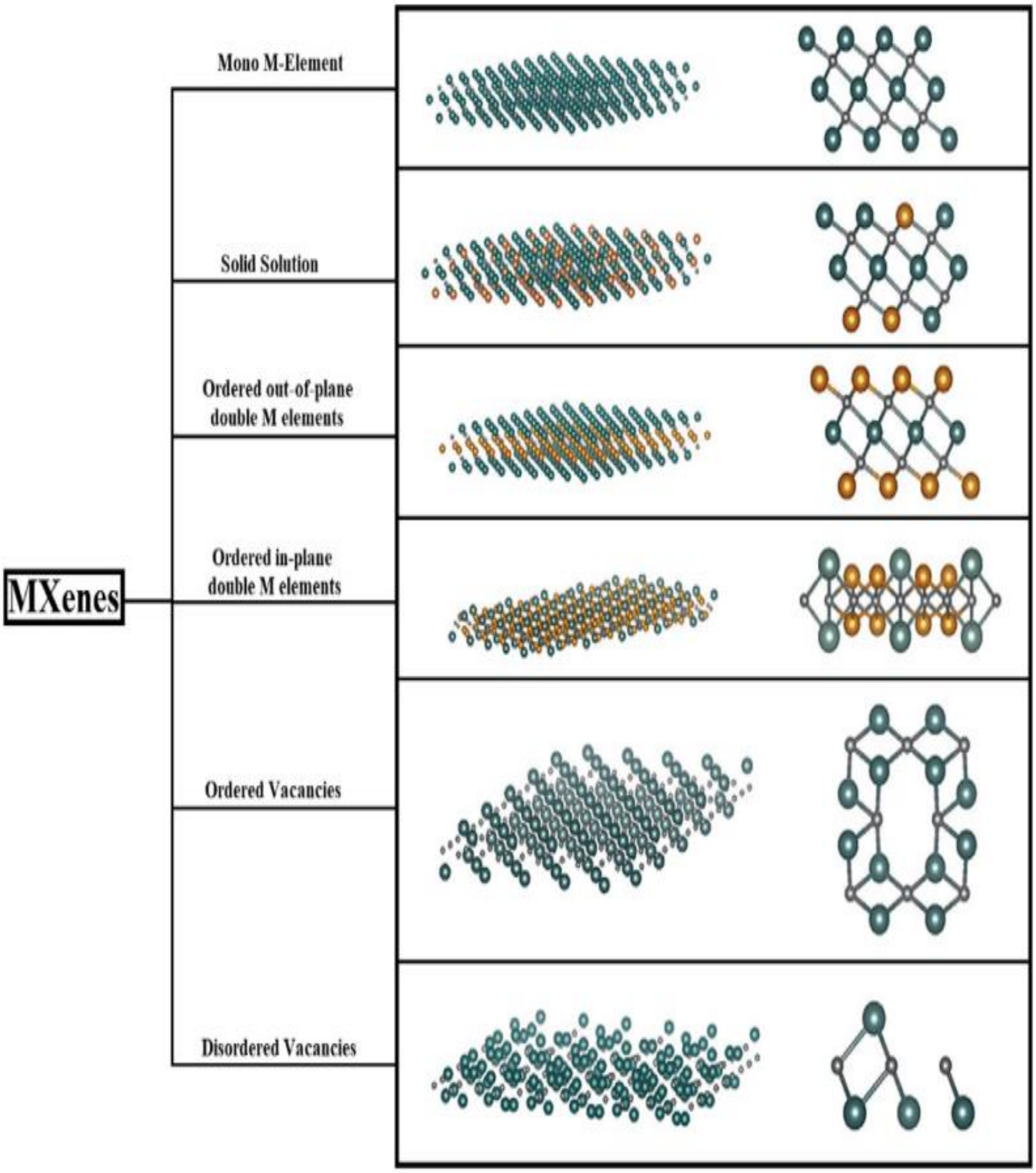


Figure 2: *MXene Structural Configurations* by Ronchi, R. M., Arantes, J. T., & Santos, S. F. (2019), diagram. Accessed June 4, 2025, from https://doi.org/10.1016/j.ceramint.2019.06.114 [63]

1.4 General Classification of MXenes

MXenes can be classified into several types based on how they are modified or combined with other materials:

1.4.1 Elemental Doped MXenes (EDMs)

Doping involves introducing atoms like nitrogen, sulfur, or phosphorus into the MXene structure. This process modifies its electrical, magnetic, and catalytic properties by altering electron density or introducing defect sites. Inspired by similar approaches in graphene research, elemental doping in MXenes improves performance in supercapacitors and batteries by enhancing conductivity, reactivity, and stability [8].

1.4.2 MXene-Based Composites (MBCs)

MXenes can also be combined with other materials to form composites. These might include polymers, carbon nanotubes, graphene, or metal oxides. The resulting composites retain the high conductivity of MXenes while gaining additional benefits like mechanical flexibility, porosity, or enhanced pseudo-capacitanceSuch materials are widely used in electrodes for lithium-ion batteries, sodium-ion batteries, and supercapacitors due to their improved electrochemical response [9].

1.5 Derivatives of Mxene:

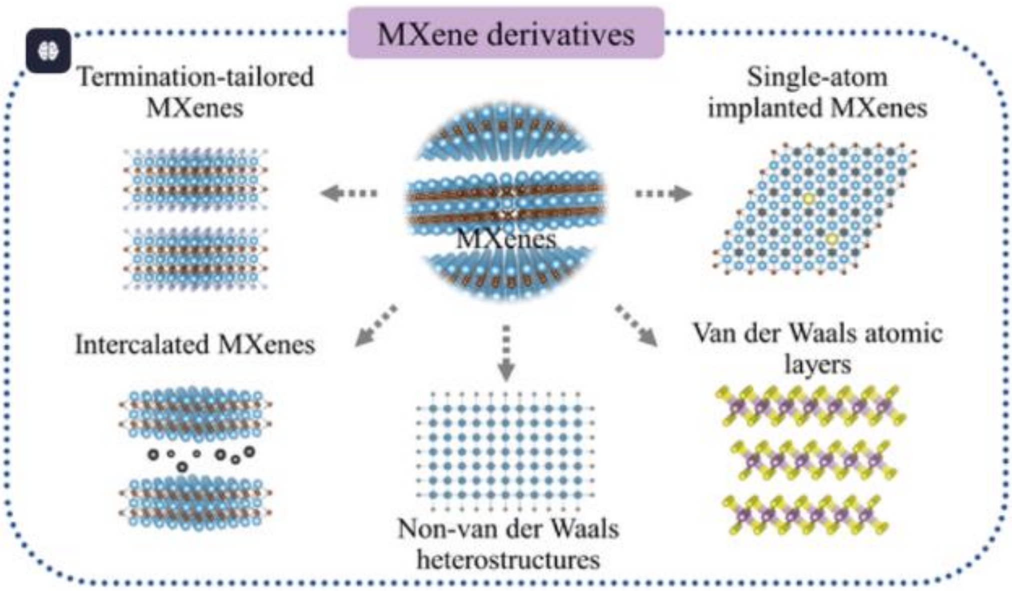


Figure 3: *MXene Derivatives for Energy Storage* by Guo, Y., Du, Z., Cao, Z., Li, B., & Yang, S. (2023), schematic. Accessed June 4, 2025, from https://doi.org/10.1002/smtd.202201559 [64]

CHAPTER 2

THEORETICAL REVIEW

METHODS OF SYNTHESIS OF MXENE

Creating MXenes typically involves breaking down layered MAX phase materials into thin, two-dimensional sheets by removing the A-layer atoms. The challenge lies in doing so without damaging the integrity of the M-X layers, which form the core of the MXene structure. There are two main approaches for synthesizing these materials: the **top-down** and **bottom-up** methods. Each has its own set of advantages depending on the desired properties and scale of production [10].

2.1. Top-down Method

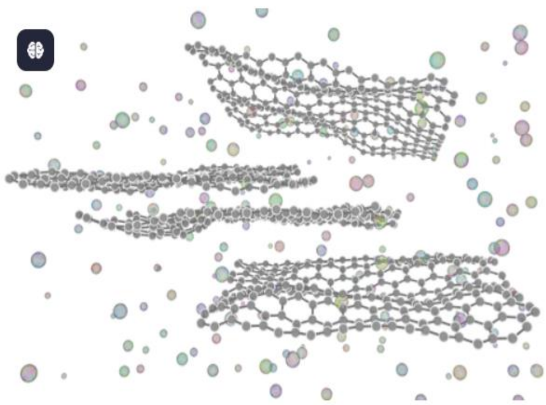
This approach starts with a bulk MAX phase and removes the A-layer atoms to form MXenes. It includes several techniques that rely on physical or chemical exfoliation [11].

2.1.1. Mechanical Exfoliation

Commonly referred to as the "Scotch tape method," this technique was first made popular through graphene research. It involves using adhesive tape to peel off thin flakes from layered materials. Although simple, it's mostly used in lab-scale experiments because it produces only small quantities and has limited control over flake size or thickness. However, the resulting MXenes are usually of high quality and have minimal defects [11].

2.1.2 Liquid Exfoliation

In this method, MAX phase powders are suspended in a solvent and subjected to ultrasound (sonication) or shear forces. This physical agitation breaks the weak van der Waals forces between the layers, separating them into thinner sheets. Sometimes, reactive ions are added to help lift the layers apart. While this method is scalable, it can result in smaller flake sizes and more structural defects compared to mechanical exfoliation [11].



Liquid exfoliation often uses bubbles to force layers apart.

Figure 4: *Liquid Exfoliation of 2D Materials* by Tahir, M., & Fatima, U. (2021), illustration. Accessed June 4, 2025, from https://doi.org/10.1002/est2.244 [65]

2.2 Bottom-up Method

Unlike the top-down approach, bottom-up synthesis involves building the material layer by layer from molecular or atomic precursors. This method is more complex but offers greater control over structure and composition [12].

2.2.1 Chemical Vapor Deposition (CVD)

CVD is a well-established method for synthesizing thin films of materials like graphene and transition metal dichalcogenides (TMDCs). It involves passing gas-phase precursors over a heated substrate where they chemically react to form a thin layer of the desired material. For MXenes, CVD is still under development, but it holds promise for producing high-purity films with minimal structural defects [12].

2.2.2 Solution-Based Synthesis

This technique uses wet-chemical reactions to form 2D materials. Examples include solvothermal methods or interface-mediated growth, where reactions are confined to the surface of a liquid. These approaches are cost-effective and scalable, though they often produce flakes with smaller lateral dimensions. Residual solvents may also affect material purity if not properly cleaned [12].

UNIQUE PROPERTIES OF MXENE

MXenes stand out among 2D materials due to their rich combination of structural, mechanical, electronic, and optical characteristics. These properties not only arise from their intrinsic atomic arrangement but are also heavily influenced by surface terminations and synthesis methods. Understanding these features is essential to tailoring MXenes for advanced energy storage systems.

3.1 Atomic Structures

The atomic arrangement of MXenes closely resembles that of other 2D materials like graphene when viewed from above. However, a side view reveals their layered nature — typically composed of a central layer of carbon or nitrogen atoms (X), sandwiched between two layers of transition metal atoms (M), forming an M–X–M structure [5].

MXenes generally adopt a hexagonal lattice. The strong covalent or metallic bonds between the M and X atoms make the core structure very stable. When surface functional groups such as –O, –OH, or –F are added during synthesis, the resulting structure becomes more stable thermodynamically compared to partially functionalized forms. These terminations also play a crucial role in defining the material's reactivity, conductivity, and overall performance [5].

3.2 Mechanical Properties

One of the most attractive features of MXenes is their excellent mechanical strength. This is primarily due to the robust bonding between the metal (M) and carbon/nitrogen (X) layers. Studies have shown that MXenes possess higher elastic moduli compared to other 2D materials like MoS_2, though slightly lower than graphene [5].

What makes MXenes especially valuable in composite applications is their strong interfacial interaction with polymers. This compatibility enhances stress transfer and overall mechanical durability, making MXenes ideal for use in flexible or wearable electronics [5].

Despite their strength, MXenes can still be engineered to maintain flexibility, particularly when formed into thin films or combined with polymeric or carbon-based matrices [5].

3.3 Electronic and Electrical Properties

The electrical behavior of MXenes is one of the key reasons for their popularity in energy-related applications. Depending on the type of surface functionalization, MXenes can behave as metallic conductors or semiconductors. For example, terminations like –O generally enhance conductivity, while –F may reduce it [5].

MXenes like $Ti_3C_2T_x$ have shown conductivity values comparable to or exceeding those of graphene oxide and carbon nanotubes. However, the number of layers, interlayer spacing, and functional groups can all impact this performance. Additionally, exposure to moisture, air, and temperature can influence their conductivity, highlighting the importance of environmental control during storage or device operation [5].

Shorter etching times and careful control of functional groups during synthesis are known to improve conductivity and reduce resistive losses in practical devices [5].

3.4 Optical Properties

MXenes also exhibit promising optical characteristics, particularly in the UV-visible and near-infrared ranges. Their optical behavior can be tuned by changing their thickness, composition, or surface chemistry. For example, $Ti_3C_2T_x$ thin films have shown high transmittance in the visible range and distinct absorption bands around 700–800 nm [13].

These properties make MXenes suitable for use in optoelectronic applications, transparent conductive films, and even in photothermal therapies. The ability to adjust the film thickness or incorporate different intercalants gives researchers flexibility in designing materials with specific optical responses [13].

3.5 Oxidative or Thermal Stability

While MXenes have many attractive features, they are somewhat sensitive to environmental exposure. In particular, titanium-based MXenes like $Ti_3C_2T_x$ tend to oxidize over time when exposed to water, air, or light. This can lead to the formation of titanium oxide, which negatively affects conductivity and structural integrity [13].

To mitigate this, researchers have explored several stabilization strategies. These include embedding MXenes in carbon coatings, storing them in inert environments, or modifying their surface terminations. Thermal treatments in controlled atmospheres (like argon) can also help preserve their structure up to high temperatures — in some cases, up to 1300 °C [13].

Understanding and improving the stability of MXenes remains a key area of research, particularly for long-term device performance and scalability [13].

IMPORTANCE OF FUNCTIONALIZATION

MXenes already exhibit outstanding conductivity and structural characteristics, but their true potential often lies in how their surfaces are chemically modified. This process — called **functionalization** — involves attaching different chemical groups or combining MXenes with other materials to improve performance in specific applications like batteries and supercapacitors [14].

Functionalization helps address several key challenges in using pristine MXenes, such as restacking of layers, limited stability in air or aqueous environments, and suboptimal electrochemical performance. By tailoring the chemistry at the atomic level, researchers can boost ion transport, increase active surface area, and improve the overall efficiency and durability of energy storage devices [14].

4.1 Enhancing Surface Reactivity and Electrochemical Behavior

One of the most direct ways to improve MXene performance is by modifying their surface terminations. During synthesis, groups like –O, –OH, or –F are naturally introduced, but they can also be adjusted afterward to fine-tune reactivity [15].

For example, –O and –OH groups make the MXene surface more hydrophilic, improving ion transport in aqueous electrolytes. This is especially useful in supercapacitors and batteries where fast ion diffusion is essential. On the other hand, doping with elements like nitrogen or sulfur can introduce new redox-active sites, leading to higher energy and power density [15].

These surface-level tweaks have a big impact on properties like charge storage capacity, electrical conductivity, and reaction kinetics [15].

4.2 Prevention of Restacking and Structural Stabilization

A common problem with layered materials like MXenes is **restacking**, where the sheets clump together due to van der Waals forces, limiting surface area and slowing down ion diffusion. Functionalization offers a way around this [16].

Introducing interlayer "pillars" — such as metal ions, organic molecules, or polymers — helps keep the layers up channels for electrolyte ions to travel through. This not only improves performance but also preserves structural stability during repeated charge/discharge cycles [16].

These modifications are especially helpful in flexible devices or when working with MXenes in slurry or ink-based formulations [16].

4.3 Improving Environmental Stability and Lifespan

MXenes, particularly those based on titanium, are prone to oxidation when exposed to air or moisture. This degrades their conductivity and weakens their structure over time. Functionalization helps protect MXenes by passivating the surface or embedding them into protective matrices [17].

For instance, coating MXene layers with polymers or incorporating them into hybrid frameworks (like graphene or carbon-based structures) can delay oxidation and extend the device's lifespan. Advanced techniques like UV-ozone treatment or chemical washing also help control surface terminations and improve resistance to degradation [17].

These enhancements are critical for real-world applications where long-term stability is a must [17].

4.4 Enabling Multi-Functional Hybrid Architectures

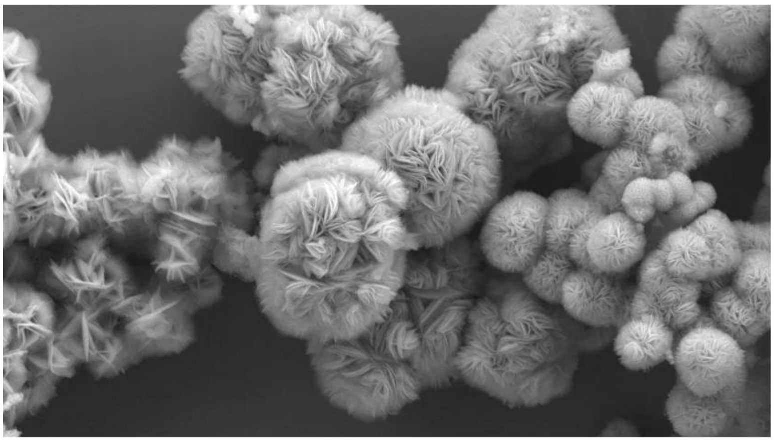
One of the most exciting outcomes of functionalization is the creation of **MXene-based hybrid materials**. By combining MXenes with other active materials — such as conductive polymers, metal oxides, or carbon nanotubes — researchers can design composites that balance high energy density, fast charge rates, and good mechanical strength [18].

These hybrid architectures are especially useful in bridging the gap between battery-like and capacitor-like behavior. For example, MXene–PANI composites combine high pseudo capacitance with flexibility, while MXene–graphene hybrids offer improved conductivity and surface area [18].

This adaptability makes MXenes suitable for everything from grid storage systems to wearable electronics [18].

CHAPTER 3

SYNTHESIS PROCESSES



SYNTHESIS OF MXENE

5.1 MXene as Precursors

The synthesis of MXenes typically begins with the bulk, layered ternary carbides or nitrides known as MAX phases. These phases are commonly identified by the chemical formula Mn+1 AXn , where M represents a transition metal (such as titanium, vanadium, chromium, niobium, or molybdenum), A is an element from Groups 13 or 14 (typically aluminum, silicon, or gallium), and X is either carbon, nitrogen, or both. The integer 'n' in the formula usually ranges from 1 to 4 [13].

MAX phases have a unique hexagonal crystal structure in which layers of Mn+1 Xn are sandwiched between single atomic layers of the A element. The M-X bonds are strong and exhibit both covalent and metallic characteristics, whereas the M-A bonds are weaker and predominantly metallic. This disparity in bond strengths makes the A element particularly vulnerable to chemical etching [13].

This layered arrangement in the MAX phases forms the foundation for the top-down approach used in MXene synthesis. The selective removal of the A element from the structure during etching is facilitated by the weaker M-A bonds, which allows the MXene layers to remain intact. Additionally, the variety of possible MAX phases—determined by variations in the M, A, and X elements—provides a broad, albeit largely unexplored, range of MXene compositions, each potentially exhibiting distinct and customizable properties suitable for diverse applications [13].

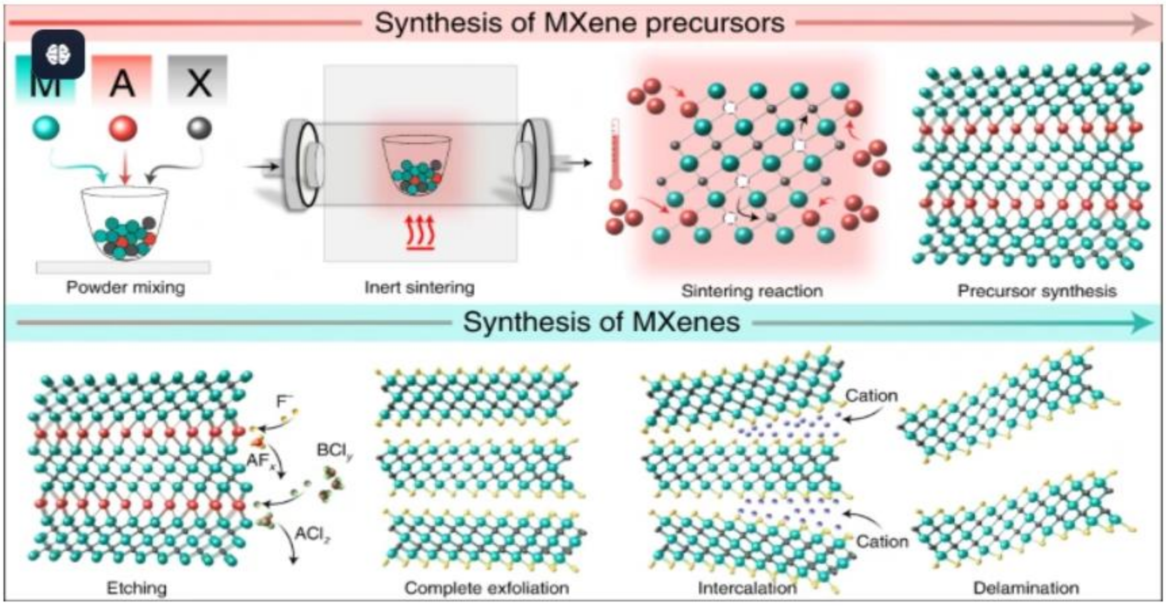


Figure 5: Synthesis of MXenes from MAX Phases by Sengupta, J., & Hussain, C. M. (2025), schematic. Accessed June 4, 2025, from https://doi.org/10.3390/bios15030127 [66]

5.1.1 Synthesis of Mxene (Ti3C2)

Typically, the synthesis of any chemical species is dependent on the strength of the bonds that must be broken in precursor or reactant molecules.
M-X and M-A bond strengths are also crucial for the development of MXenes. M-A-A-A– bonds are more easily broken and weaker than M–X bonds (Khazaei et al. 2013, 2018; Zhou et al. 2017). Consequently, the A-layer from the original or parent MAX phase is removed during the room temperature etching procedure to create MXenes [19]. Parent MAX phase through the room-temperature etching procedure. For this purpose, etching agents that contain aqueous fluorides are frequently used. In short, the aqueous HF and MAX phase powders in this procedure are agitated at room temperature for a certain amount of time, which causes weak interactions with fluorides, oxygen, or hydroxyl groups to replace the strong bonds between the MX-layer and A-layer. Following stirring, the contents are filtered or centrifuged to separate the solid from the supernatant. The solid is then washed with distilled or deionized water to bring the mixture's pH down to a value between 4 and 6 [19].

$$Ti_3AlC_2 + 3HF \rightarrow Ti_3C_2 + AlF_3 + 3/2\ H_2$$
$$Ti_3C_2 + 2H_2O \rightarrow Ti_3C_2\,(OH)_2 + H_2$$
$$Ti_3C_2 + 2HF \rightarrow Ti_3C_2F_2 + H_2$$

The creation of Ti3C2 (OH)2 MXenes and Ti3C2F2 by treating MAX phase Ti3AlC2 with aqueous HF can be summed up as follows: Etching the MAX phase at high temperatures is another method for creating MXenes. This method developed an MXene based on nitride. Ti4AlN3's Al layer, a fused fluoride salt mixture of LiF (29%),
(12%), and KF (59%), was employed to etch the powder [20].

Mono-layered Ti4N3Tx is formed during further treatment with tetrabutylammonium hydroxide (TBAOH). There have also been reports of certain exceptionally high-temperature etching techniques, such as the removal of the Si-layer from Ti3SiC2 using fused cryolite at 960 °C and the etching of In-layer Ti2 InC by sublimating In-layers at a temperature of roughly 800 °C. Nevertheless, the structures that were produced were not entirely two-dimensional. Fig. 1 displays a diagrammatic representation of the several processes involved in the synthesis of MXenes [21].

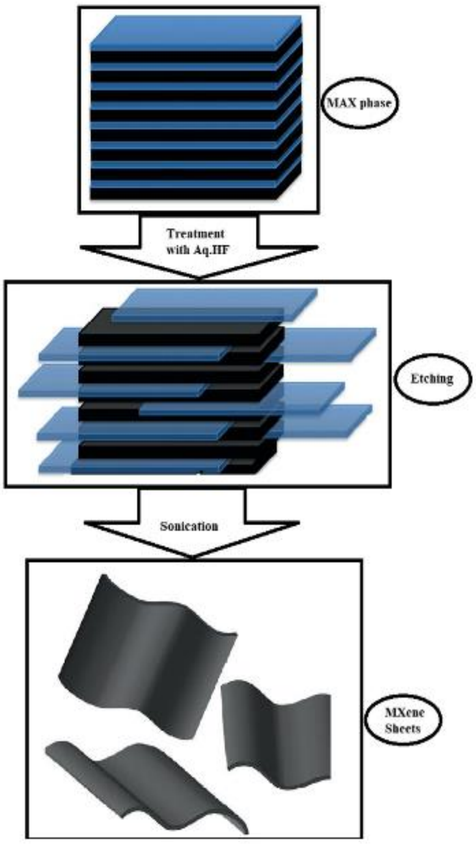


Figure 6: *MXene/MOF Composites as Photocatalysts* by Fattah-alhosseini, A., Sangarimotlagh, Z., Karbasi, M., & Kaseem, M. (2024), schematic. Accessed June 4, 2025, from https://doi.org/10.1016/j.nanoso.2024.101192 [67]

5.2 Etching Techniques for MXene Synthesis

Etching is a critical step in the top-down synthesis of MXenes, determining both the success of the process and the properties of the final material. The choice of etchant influences the efficiency and safety of the process and also governs the surface terminations of the resulting MXenes, which in turn affect their chemical and physical properties [22].

5.2.1 Hydrofluoric Acid (HF) Etching

HF etching was the first method used to synthesize MXenes and remains one of the most effective. In this approach, MAX powders are immersed in concentrated HF (typically 48%) at room temperature for several hours. The etching process removes the A element, forming soluble aluminum fluoride (AlF_3), and hydrogen gas is released as a byproduct. Additionally, surface terminations such as –F, –OH, and –O are introduced to the MXene [23].

While HF etching is highly effective, it comes with significant safety concerns due to the toxic and corrosive nature of HF. It also presents environmental challenges and is difficult to scale for industrial use. Despite these issues, HF etching remains a benchmark method due to its reliability in producing high-quality MXenes with a well-defined layered structure [23].

5.2.2 In-situ HF Etching (LiF + HCl)

To address the hazards associated with HF, an alternative method using lithium fluoride (LiF) and hydrochloric acid (HCl) has been developed. This approach generates HF in situ, providing a safer and more controlled etching environment. In this method, Li^+ ions are intercalated between the MXene layers during the reaction, promoting the delamination of the layers into fewer or even monolayer flakes [24].

While this method is safer and more environmentally friendly, it requires longer etching times compared to using concentrated HF, and thorough washing is necessary to remove residual salts that may affect the material's properties [24].

5.2.3 Molten Salt Etching

Molten salt etching is a newer approach that involves treating MAX phases with halide salt mixtures (such as $ZnCl_2$, $CuCl_2$, or $FeCl_3$) at elevated temperatures (400–600°C). This technique does not rely on acids, making it a more environmentally friendly option. In this process, metal halides react with the A-layer elements to form stable metal chlorides, while the MXene surface is terminated with halides or oxides [25].

Molten salt etching eliminates the need for toxic HF and offers the potential for creating MXenes with unique surface terminations and enhanced thermal stability. However, this method requires specialized equipment and high temperatures, and the resulting surface terminations may vary, affecting the compatibility of the MXenes for certain applications [25].

These etching methods exemplify the evolution of MXene synthesis from **hazardous chemical treatments** to **eco-friendly, tunable, and scalable processes**. The selected route greatly impacts not just the safety and yield, but also the **electrochemical properties**, **interlayer spacing**, and **functional group diversity**—all of which are critical for tailoring MXenes for specific energy storage applications [26].

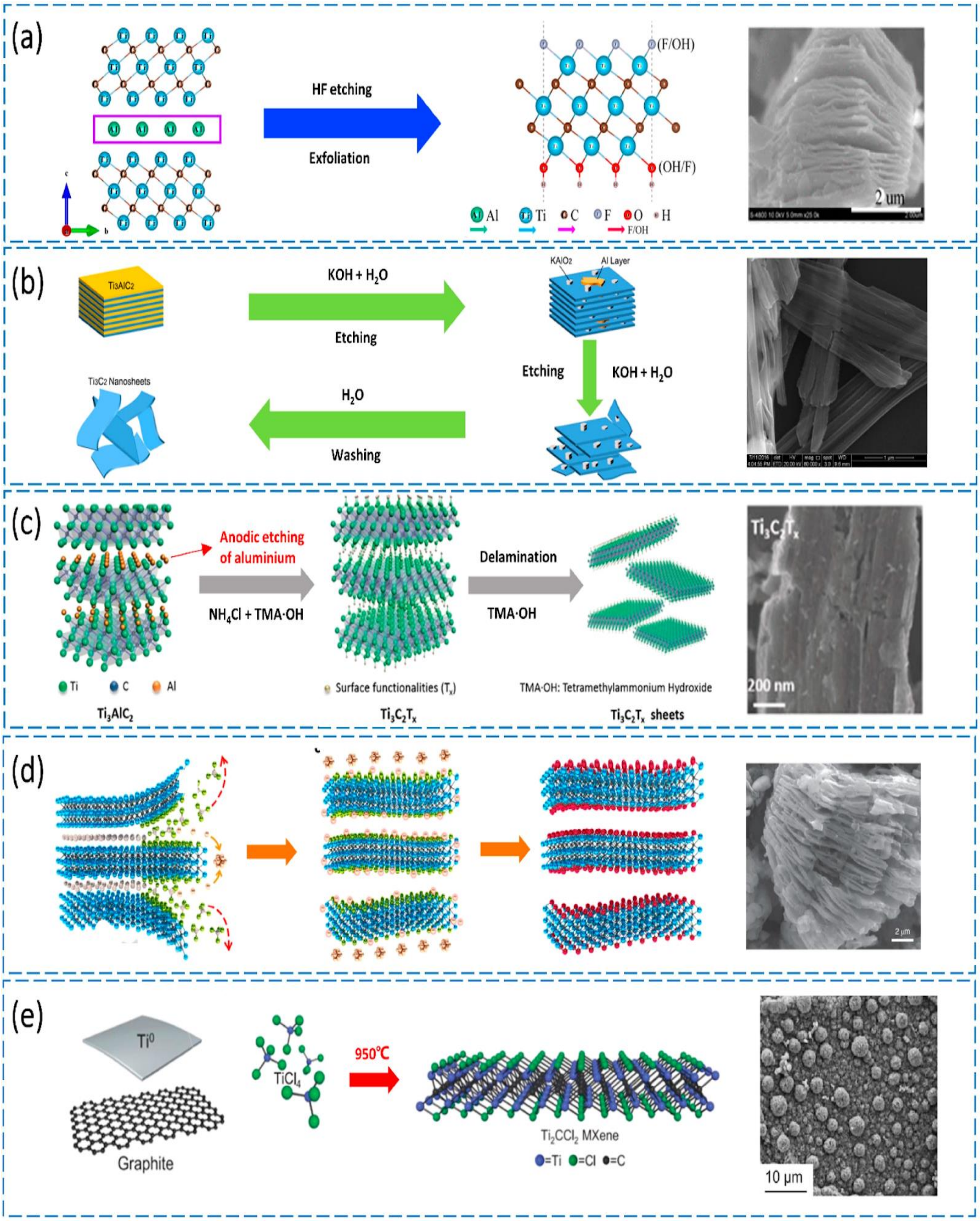
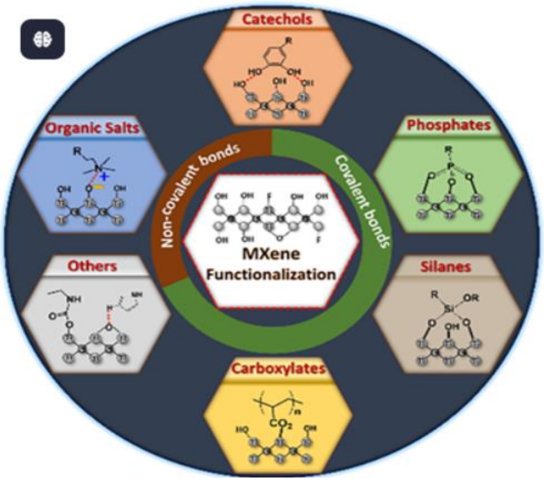


Figure 7: *MXenes for Sustainable Energy Applications* by Jussambayev, M., Shakenov, K., Sultakhan, S., Zhantikeyev, U., Askaruly, K., Toshtay, K., & Azat, S. (2025), diagram. Accessed June 4, 2025, from https://doi.org/10.1016/j.cartre.2025.100471 [68]

CHAPTER 4
FUNCTIONALIZATION TECHNIQUES



FUNCTIONALIZATION TECHNIQUES

Although MXenes possess excellent intrinsic properties, their full potential is realized through functionalization, a process that introduces surface groups, dopants, or hybrid materials to tailor the MXenes for specific applications. Functionalization can enhance the material's conductivity, electrochemical performance, and mechanical properties. The following sections discuss the primary strategies used to functionalize MXenes [27].

6.1 Surface Termination Engineering

Surface terminations, which are typically introduced during the etching process, play a crucial role in the properties of MXenes. Common surface groups include –OH, –F, and –O, and each group impacts the material in different ways. For example, –O terminations increase the material's metallic conductivity and facilitate redox reactions, while –OH groups enhance hydrophilicity and ion intercalation. On the other hand, F groups typically have lower conductivity and can inhibit certain electrochemical processes [28].

Controlling the type and ratio of these terminations is essential for optimizing MXenes for specific applications. Post-synthesis treatments, such as annealing in reducing atmospheres, chemical washing, and UV-ozone treatment, can be employed to modify the surface terminations, removing –F groups and increasing the abundance of –O or –OH groups [28].

6.2 Heteroatom Doping

Doping MXenes with heteroatoms like nitrogen (N), sulfur (S), boron (B), or phosphorus (P) is another strategy to enhance their performance. These dopants can modify the electronic structure of MXenes, improving properties such as conductivity and electrocatalytic activity. Nitrogen doping, for example, introduces electron-donating sites that enhance electronic conductivity and charge carrier mobility, making it particularly useful for energy storage applications [29].

Other dopants, such as sulfur, phosphorus, and boron, can further improve the electrochemical behavior of MXenes. Sulfur doping, for instance, can enhance catalytic performance for metal-sulfur batteries, while phosphorus doping creates active sites for pseudo-capacitance. These doping processes are typically conducted through thermal treatments or chemical methods [29].

6.3 Interlayer Engineering and Pillaring

One of the challenges in using MXenes is the tendency for the layers to restack, which reduces the available surface area and hampers ion diffusion. To address this issue, interlayer engineering is employed to introduce spacers or pillars between the layers. These spacers can be organic molecules, metal cations, or polymers, which not only expand the interlayer spacing but also enhance the material's mechanical flexibility and electrochemical functionality [30].

6.4 Hybridization with Other Materials

MXenes can also be combined with other materials to form hybrid composites that leverage the strengths of both components. For example, MXene-carbon hybrids, which integrate graphene or carbon nanotubes, can improve conductivity and mechanical strength. Similarly, MXene-polymer composites, such as those incorporating polyaniline or polypyrrole, offer high pseudocapacitance and flexibility, making them ideal for wearable energy storage devices. MXene-metal oxide hybrids, incorporating materials like MnO_2 or TiO_2, improve cycling performance and structural stability, enhancing the overall durability of the devices [31].

By customizing these functionalization strategies, MXenes can be tailored for a wide range of applications in energy storage, catalysis, and beyond [31].

Supercapacitors, sometimes referred to as electrochemical capacitors, represent a unique class of energy storage systems that sit between traditional capacitors and batteries. They are renowned for their high power output, excellent cycle life, and fast charge-discharge response. Yet, many of the conventional materials used in their electrodes, especially carbon-based ones like activated carbon and graphene, are limited in energy density because they lack redox-active surfaces [31].

MXenes, with their inherently layered two-dimensional structure and high electronic conductivity, offer a compelling alternative. Their surfaces are rich in functional terminations capable of engaging in redox reactions, while the interlayer galleries support efficient ion transport. This dual mechanism—combining electric double-layer capacitance and pseudocapacitance—makes MXenes particularly attractive for advanced supercapacitor electrodes [31].

Among the various MXenes explored, $\mathbf{Ti_3C_2T_x}$ has garnered the most attention for supercapacitor applications. Studies have repeatedly highlighted its ability to deliver exceptional capacitance and maintain stability over prolonged cycles. For instance, in an alkaline electrolyte like KOH, this material has demonstrated a volumetric capacitance as high as 340 F/cm^3, with full retention even after 10,000 charge-discharge cycles. In another setup using 1 M H_2SO_4, $Ti_3C_2T_x$ exhibited an impressive 900 F/cm^3, marking one of the best performances among known materials [32].

Such outstanding capabilities arise from several factors: the rapid movement of ions within its hydrated interlayers, surface redox activity involving hydroxyl and oxygen groups, and a well-connected internal network that supports electronic conductivity. Research has also shown that the synthesis method, including the type of etchant and intercalating ion (Li^+, K^+, NH_4^+, etc.), can significantly influence $Ti_3C_2T_x$'s capacitive performance [32].

CHAPTER 5

ENERGY STORAGE APPLICATIONS

MXENE-BASED SUPERCAPACITORS

7.1 Electric Double Layer Capacitors (EDLCs)

MXenes function effectively as EDLC materials, where energy is stored via electrostatic interactions at the interface of the electrode and electrolyte. Their intrinsic high surface area and layered configuration allow ions to quickly access and fill these spaces, while their metallic conductivity ensures seamless electron transport [33].

7.2 Pseudo capacitors

Many MXenes also exhibit pseudocapacitance due to surface functionalities like –O and –OH groups, which participate in reversible Faradaic reactions. This additional redox mechanism enables greater energy storage, especially in electrolytes that are either acidic or neutral [33].

7.3 Hybrid Supercapacitors

To enhance both power and energy density, researchers have employed MXenes in asymmetric or hybrid capacitor architectures. Typically, this involves pairing a pseudocapacitive MXene with an EDLC-type material such as activated carbon, leveraging the strengths of both components [33].

7.4 MXene Composites for Supercapacitor Applications

One of the main challenges with MXenes is the tendency of their layers to restack, which reduces ion accessibility and narrows the electrochemical window. To address this, composite materials combining MXenes with carbonaceous structures, polymers, or metal oxides have been engineered [34].

7.4.1 $Ti_3C_2T_x$/Graphene and rGO Hybrids

By integrating MXenes with graphene derivatives like reduced graphene oxide (rGO), researchers have produced flexible films that exhibit superior surface area and structural integrity. These hybrids have demonstrated specific capacitances around 335 F/g at a scan rate of 5 mV/s while maintaining complete capacitance even after 20,000 cycles. Self-assembly techniques have enabled the creation of highly porous, mechanically stable films that enhance ion mobility [35].

7.4.2 MXene/Polyaniline Composites

Conductive polymers such as polyaniline (PANI) are often combined with MXenes to amplify their pseudocapacitive behavior. One such MXene@PANI system achieved a volumetric capacitance of 1632 F/cm^3 and a specific capacitance of 510 F/g, with excellent cycling stability

(~85% retention over 20,000 cycles). The presence of amine functionalities from PANI promotes strong redox activity and improves the overall mechanical durability of the electrode [36].

7.4.3 MXene–Carbon Nanotube and Mesoporous Carbon Hybrids

MXene-CNT composites, particularly in the form of aerogels, offer excellent electrochemical performance in organic electrolytes. These structures prevent restacking, maintain conductivity, and support efficient ion flow, even at high scan rates [37]. Similarly, MXene hybrids with ordered mesoporous carbon (OMC) and acid-activated carbon (AAC) have achieved capacitances over 800 F/cm³ in 3 M KOH, alongside long operational lifetimes exceeding 100,000 cycles [37].

7.5 Asymmetric Supercapacitor Designs

To expand voltage windows and enhance energy density, MXenes have been explored in asymmetric supercapacitor configurations. One example includes a composite of molybdenum- and titanium-based MXenes (3Mo:1Ti) as the negative electrode, paired with activated carbon as the positive counterpart [38]. This system achieved a cell voltage of around 1.5 V, a discharge capacitance of up to 50 F/g at 5 A/g, and maintained 75% of its initial performance after prolonged cycling. The improved voltage range directly contributes to a higher energy density, in the range of 16–18 Wh/kg [38].

7.6 Effect of Functionalization on Supercapacitor Performance

Functionalization is key to tailoring MXene properties for enhanced electrochemical performance. Expanding the interlayer spacing through organic molecules or ionic species can significantly boost ion accessibility and diffusion. Surface doping, particularly with elements like nitrogen or sulfur, has been shown to elevate redox activity and overall capacitance [39].

Moreover, post-synthesis treatments such as exposure to alkaline media or thermal processing help fine-tune the surface terminations. For instance, glycine-treated $Ti_3C_2T_x$ not only improved interlayer spacing but also introduced Ti–N bonding, which positively influenced charge storage characteristics [39].

MXENE-BASED BATTERIES

Rechargeable batteries have become an integral part of modern life, driving everything from handheld gadgets to electric vehicles. While lithium-ion batteries (LIBs) continue to lead in terms of energy density and durability, the growing demand for cleaner, more efficient technologies has fueled exploration into alternative systems like sodium-ion batteries (SIBs). Central to the advancement of these batteries is the search for innovative electrode materials that can enhance both performance and sustainability [40].

MXenes have recently garnered attention for their potential in battery applications. Thanks to their unique two-dimensional morphology, high electrical conductivity, and customizable surface chemistry, they show strong potential for storing various metal ions, including lithium, sodium, potassium, and magnesium. The ability of MXenes to intercalate these ions within their layered structures makes them a highly versatile platform for next-generation energy storage devices [40].

8.1 MXenes in Lithium-Ion Batteries

8.1.1 Mechanism and Performance

MXenes like $Ti_3C_2T_x$ are particularly well-suited for LIB applications due to their fast ion transport and structural adaptability. When lithium ions are introduced, they intercalate between the layers of the MXene, where surface terminations such as –O and –OH actively participate in redox processes to store charge. This combination results in efficient energy storage and rapid cycling [41].

Laboratory investigations have reported that $Ti_3C_2T_x$ can theoretically offer up to 320 mAh g^{11} in capacity. Moreover, when MXenes are engineered into composites, such as those with red phosphorus or certain metal oxides, capacities can exceed 1000 mAh· g^{1}, revealing their promise for high-performance LIBs [41].

In lithium-sulfur batteries, MXenes also demonstrate the ability to limit the "shuttle effect," a major issue where soluble polysulfides migrate during cycling. By chemically binding to these species, MXenes help stabilize sulfur utilization and improve long-term cycling behavior [41].

8.1.2 Composite Strategies for LIBs

To further enhance performance, researchers have developed MXene-based hybrids that combine different materials for synergistic effects. One strategy involves blending $Ti_3C_2T_x$ with red phosphorus through ball milling, resulting in a nanostructured material capable of delivering up to 1112 mAh· g^{1}. Other approaches involve integrating metal oxides like VO_2 or

$CoNiO_2$, which not only boost capacity but also improve rate capabilities and durability under rapid charging [42].

Conductive polymers like polyaniline (PANI) are also used to coat or embed MXenes. These polymers add flexibility, increase electrical conductivity, and support structural integrity during charge-discharge cycles, effectively addressing volume expansion and other degradation mechanisms [42].

8.2 MXenes in Sodium-Ion Batteries (SIBs)

8.2.1 Motivation and Challenges

Sodium-ion batteries have emerged as an appealing alternative to lithium-based systems, largely due to the widespread availability and lower cost of sodium. However, the larger size of sodium ions compared to lithium poses a challenge. This size difference makes it difficult to identify electrode materials that offer both structural stability and high sodium storage capacity [43].

To be effective in SIBs, electrode materials must have wide interlayer spacing to support ion diffusion, structural resilience to accommodate volume changes and strong electronic conductivity. MXenes meet all of these criteria, making them a highly promising candidates in the field [43].

8.2.2 Sodium Storage Using Pure MXenes

Initial experiments using pure $Ti_3C_2T_x$ as an anode material faced issues related to restacking. The layers tended to collapse due to van der Waals forces, limiting access to sodium ions. To overcome this, researchers began creating three-dimensional porous MXene networks. By using fabrication techniques like additive manufacturing and sacrificial templating—often involving nitrogen-rich polymers—they were able to widen the interlayer distance and create more accessible ion pathways [44].

For example, Fan and colleagues developed a 3D nitrogen-doped MXene electrode using melamine-formaldehyde nanospheres. This structure not only enhanced ion diffusion but also improved redox reactivity and cycling stability [44].

8.2.3 MXene-Based Composites for Sodium Storage

To further enhance the performance of MXene electrodes in SIBs, various composites have been developed [45].

8.2.3.1 Sulfide-Based Composites:

Integrating MXenes with metal sulfides like SnS or MoS_2 has proven effective. In one configuration, SnS combined with $Ti_3C_2T_x$ improved both surface area and interlayer spacing, resulting in a capacity of nearly 256 mAh· g^1 at a current density of 1000 mA· g^1. Another design involving MoS_2 layered with MXene and carbon yielded exceptional cycling stability, retaining almost 95% of capacity after 3000 cycles at high current rates [45].

8.2.3.2 Oxide-Based Composites:

MXene-oxide hybrids, such as Sb_2O_3 anchored onto $Ti_3C_2T_x$, have demonstrated robust sodium storage with capacities reaching 472 mAh· g^1 over 100 cycles. These materials offer enhanced electron transport and facilitate ion diffusion, both of which are vital for high-rate performance [46].

8.2.3.3 Carbon Nanotube-Based Composites:

Blending MXenes with carbon nanotubes creates flexible, conductive films ideal for sodium storage. Such configurations have shown volumetric capacities of 421 mAh· cm^3 at slower rates and maintained 89 mAh· cm^3 even at rapid discharge rates (5000 mA· g^1). These hybrids capitalize on the structural resilience and conductivity of carbon to stabilize MXene layers during operation [46].

8.2.4 Functionalization Strategies to Enhance SIB Performance

Functionalization plays a key role in optimizing MXene electrodes for sodium-ion batteries. By expanding the interlayer distance through organic molecules or ionic intercalation, researchers improve ion accessibility. Adding polar groups such as –O or –S enhances redox activity and overall charge storage [47]. Another effective approach involves preventing layer restacking. For instance, Zhao et al. modified $Ti_3C_2T_x$ using NaOH, producing a sodium-functionalized MXene (Na-c-$Ti_3C_2T_x$) that achieved a capacity of 130 mAh· g^1 after 500 cycles, outperforming similar modifications done with LiOH or TBAOH. These treatments help maintain a stable structure and improve long-term cycling behavior [47].

MXENE-BASED HYBRID ENERGY STORAGE DEVICES

Hybrid energy storage systems (HESDs) are designed to bridge the performance gap between batteries and supercapacitors. While batteries offer high energy storage capacity, supercapacitors are known for their rapid charge and discharge rates. By combining the strengths of both, HESDs can deliver a balanced performance, offering both an extended energy supply and fast responsiveness [48].

MXenes are particularly promising in this context due to their hybrid characteristics. These 2D materials exhibit both capacitive and pseudocapacitive behavior [48]. Their metallic-level conductivity, layered structure, and surface functional groups allow for efficient charge transport and fast ion diffusion. Moreover, MXenes readily interact with other materials, making them ideal for designing multifunctional electrodes in hybrid devices [48].

9.1 Structural Requirements for Hybrid Devices

A well-designed hybrid energy storage device requires a combination of electrode types. Typically, one electrode behaves like a battery—storing charge through redox reactions—while the other functions more like a capacitor, storing charge electrostatically. This setup allows for both high energy and power outputs [49].

To achieve stable and efficient performance, the system should include:

- A Faradaic (battery-type) electrode, is ten based on materials like transition metal oxides or red phosphorus [49].
- A capacitive electrode—frequently composed of carbon or MXenesdeliversr fast charge-discharge capabilities [49].
- A wide operational voltage range and a compatible, stable electrolyte [49].
- A structure that is both mechanically strong and chemically stable [49].

MXenes contribute to these systems in several roles: they enhance conductivity, offer platforms for ion intercalation, and reinforce the mechanical framework, especially when integrated with porous or other 2D materials [49].

9.2 MXene–Carbon Hybrids for Hybrid Devices

One of the most explored routes to creating high-performance hybrid supercapacitors involves combining MXenes with carbon-based materials. For instance, $Ti_3C_2T_x$ blended with reduced graphene oxide (rGO) results in a flexible, conductive material that performs exceptionally well in asymmetric configurations [50].

Micro-ultracapacitors made with this composite have demonstrated strong electrochemical behavior, outperforming many traditional graphene-only devices. These hybrid films not only enhance charge storage but also maintain structural integrity, especially in solid-state formats. The synergy between MXene and rGO arises from their complementary properties and is further improved through methods like electrostatic self-assembly, which fosters better ion diffusion and structural cohesion [50].

9.3 MXene–Metal Oxide Hybrids

Pairing MXenes with metal oxides opens new possibilities for hybrid energy storage. These composites often combine the fast kinetics of supercapacitors with the deeper charge storage of battery-type materials [51].

One notable example involves 3D aerogels composed of Co_3O_4, MXene, and rGO. These structures offer high porosity and interconnected channels that enhance ion transport and electrochemical accessibility. Such configurations have demonstrated impressive capacitance values and remain stable over extensive cycling, making them suitable for long-term applications [51].

9.4 MXene–Polymer Hybrids in Flexible Devices

Conductive polymers like polyaniline (PANI) and polypyrrole (PPy) have been successfully combined with MXenes to produce highly flexible and efficient energy storage materials. The polymers contribute redox activity, while the MXene layers improve conductivity and offer structural support [52]. Composites such as MXene@PANI have shown remarkable performance in terms of specific and volumetric capacitance. More importantly, they exhibit excellent cycle stability over tens of thousands of charge-discharge cycles. These features make them particularly attractive for integration into wearable electronics, where flexibility and durability are essential [52].

9.5 Solid-State and Printable Hybrid Systems

The evolution of energy storage is moving beyond traditional liquid-electrolyte systems, with solid-state and printable devices gaining traction. MXenes, due to their hydrophilic nature and excellent dispersion in aqueous media, are particularly compatible with such innovations [53].

Printed MXene–graphene electrodes, for instance, offer a high degree of flexibility and mechanical strength without sacrificing electrochemical performance. Meanwhile, MXene-based hydrogels and aerogels are finding roles in soft robotics and wearable technologies, thanks to their stretchable and shape-conforming properties. Using polymer-based electrolytes like PVA-H_3PO_4 further enables integration into flexible platforms without hindering ion conductivity [53].

9.6 Asymmetric and Symmetric Hybrid Configurations

In asymmetric setups, MXene typically serves as the negative electrode, paired with a high-voltage material such as activated carbon or MnO_2 as the positive electrode. One such configuration using $Ti_3C_2T_x$ and activated carbon achieved a cell voltage of 1.5 V and delivered solid energy density and capacitance, with consistent performance over extended use [54].

Symmetric configurations, where MXene is used for both electrodes, also show great promise. These systems capitalize on MXene's ambipolar charge storage behavior and robust structure, leading to high volumetric energy densities and excellent cycle life. Such devices are not only efficient but also compact and well-suited for integrated energy systems [54].

9.7 MXenes in Sodium-Ion Hybrid Capacitors (SIHCs)

An exciting direction in MXene research involves their application in sodium-ion hybrid capacitors. These devices combine a sodium-intercalating anode with a fast-response supercapacitor-type cathode, aiming to strike a balance between high energy and power [55].

A notable development by Fan and co-workers involved the use of 3D-printed, nitrogen-doped MXene scaffolds for SIHCs. These structures exhibited excellent high-rate performance and mechanical durability. The additive manufacturing approach also indicates strong potential for scalability, pointing to a future where MXene-based SIHCs could become a low-cost and efficient solution for next-generation energy storage [55].

CHAPTER 6

RECENT TRENDS AND DEVELOPMENTS IN MXENE-BASED ENERGY STORAGE APPLICATIONS

RECENT TRENDS AND DEVELOPMENTS IN MXENE-BASED ENERGY STORAGE APPLICATIONS

10.1 Diversification Beyond $Ti_3C_2T_x$: The Rise of Non-Ti MXenes

- Since $Ti_3C_2T_x$ was first introduced in 2011, it has remained the focal point of MXene research, largely due to its ease of production and reliable electrochemical performance. However, the landscape is rapidly evolving, with increasing interest in non-titanium-based MXenes such as V_2CT_x, Nb_2CT_x, Mo_2CT_x, and Cr_2CT_x. These alternatives, often in the M_2X or M_3X_2 phase families, offer distinct advantages including tunable electronic structures, improved resistance to oxidation, and even higher theoretical charge storage capacities [56].
- A noteworthy example is Nb_2CT_x, which has exhibited specific capacitances surpassing 1800 F/g along with impressive cycle stability, retaining over 90% capacity even after 10,000 cycles. This level of performance, especially in aqueous and gel polymer electrolytes, positions non-Ti MXenes as highly promising candidates for next-generation supercapacitors and hybrid storage systems [57].

10.2 M_4X_3 MXenes: A New Frontier

- The discovery and exploration of M_4X_3-type MXenes, such as $Nb_4C_3T_x$ and $V_4C_3T_x$, mark a significant advancement in the field. These materials possess broader interlayer spacing, which facilitates better ion transport and enhances thermal and structural stability—critical features for use in demanding environments like lithium-ion batteries [58].
- Among the emerging M_4X_3 class, $V_4C_3T_x$ has shown a clear edge over $Ti_3C_2T_x$ in terms of specific capacitance, cycle stability, and resilience against structural fatigue. Similarly, $Nb_4C_3T_x$ demonstrates efficient lithium-ion intercalation, making it an attractive candidate for battery anodes. While synthesizing these compounds remains complex and resource-intensive, their unique physicochemical features continue to draw significant research attention [58].

10.3 Doping and Functionalization Strategies: Boosting Electrochemical Efficiency

- One of the major trends in MXene research is the use of doping and surface modification techniques to enhance their electrochemical performance. Doping elements like nickel into Nb_2CT_x, for instance, has been shown to increase surface area and electrical conductivity, thereby improving both capacitance and durability [59].

- Advanced computational studies further suggest that defect engineering and functionalization can elevate the theoretical capacitance of MXenes well beyond that of conventional materials. Additionally, post-synthesis treatments such as UV exposure, alkali washing, and nitrogen doping have proven effective in tuning surface chemistry and expanding the interlayer gaps, allowing for improved ion transport and charge storage [59].

10.4 Integration into Hybrid and Solid-State Systems

- MXenes are increasingly being integrated into hybrid and solid-state devices as the field moves away from traditional liquid electrolyte systems. Recent developments in printable MXene–graphene inks have enabled the fabrication of flexible, miniaturized supercapacitors tailored for wearable and on-body electronics [60].
- Moreover, the utility of non-Ti MXenes like V_2CT_x has been demonstrated in sodium-ion hybrid capacitors, where they function effectively as positive electrodes. These systems combine the high energy density of batteries with the fast response of capacitors, achieving capacities near 70 mAh/g with commendable stability [60].
- Hybrid architectures that incorporate conductive polymers or transition metal oxides further expand the versatility of MXenes. These combinations leverage both the capacitive and redox-active characteristics of the components, pushing the boundaries of achievable energy and power densities [60].

10.5 Bottom-Up Synthesis and Sustainable Processing

- As researchers seek safer and more sustainable ways to produce MXenes, several innovative bottom-up approaches are being explored. Traditional methods, like HF-based etching, pose environmental and safety challenges. In response, techniques such as molten salt etching, electrochemical methods, and template-assisted chemical vapor deposition (CVD) are gaining popularity [61].
- These alternative routes not only reduce hazards but also enable greater control over surface terminations and structural features. For example, molten salt routes yield high-quality, defect-free flakes with unique surface properties, while electrochemical etching allows precise tuning of functional groups [61]. Template-assisted CVD, on the other hand, opens the door to architected MXenes with vertical alignment or atomically ordered structures. These advances also support the synthesis of novel variants like high-entropy MXenes, offering fresh possibilities in energy storage design [61].

10.6 Unresolved Challenges and Future Opportunities

- Despite the impressive strides made so far, several challenges continue to limit the widespread application of MXenes. One major concern is their susceptibility to oxidation under ambient conditions, which affects both stability and performance. In addition, the tendency of MXene sheets to restack reduces ion accessibility and hinders effective transport, posing obstacles to scalability and real-world device integration [62].
- Efforts to address these issues are underway, with researchers exploring new composite structures, exfoliation strategies, and environmental stabilization techniques. The future of MXene research is likely to be driven by data-guided approaches, including machine learning algorithms for materials discovery and design [62]. Other emerging directions include the development of high-entropy MXenes, advanced doping techniques, and the creation of bio-inspired architectures. Tailoring MXenes for specific applications—ranging from electric vehicles to microelectronics—will be key to unlocking their full potential in energy storage [62].

CHAPTER 7

CONCLUSION

CONCLUSION

This review has highlighted the growing prominence of MXenes as versatile materials for energy storage applications. Through controlled synthesis and strategic functionalization, MXenes exhibit exceptional electrochemical performance across supercapacitors, lithium-ion, and sodium-ion batteries, as well as hybrid storage systems. Their unique combination of metallic conductivity, layered structure, and surface reactivity makes them ideal candidates for high-rate and high-capacity energy devices. Functionalization, in particular, plays a vital role in enhancing ion accessibility, electrical conductivity, and device stability. Recent advancements in synthesis methods—ranging from fluoride-free etching to bottom-up and molten salt approaches—have enabled the development of MXenes with improved structural integrity and tunable properties. The integration of MXenes into hybrid architectures and solid-state configurations has further expanded their applicability in flexible and wearable technologies. However, challenges such as environmental stability, restacking, and large-scale production still limit their widespread implementation. Continued efforts in material design, functional tuning, and sustainable processing are essential to fully realize MXenes' potential in the next generation of high-performance, scalable, and environmentally responsible energy storage systems.

CHAPTER 8

REFERENCES

REFERENCES

[1] Ibrahim Khan, Khalid Saeed, Idrees Khan, Nanoparticles: Properties, applications and toxicities, Arabian Journal of Chemistry, Volume 12, Issue 7, 2019, Pages 908-931, ISSN 1878-5352

[2] Altammar KA. A review on nanoparticles: characteristics, synthesis, applications, and challenges. Front Microbiol. 2023 Apr 17;14:1155622. doi: 10.3389/fmicb.2023.1155622. PMID: 37180257; PMCID: PMC10168541.

[3] Nadeem Baig, Two-dimensional nanomaterials: A critical review of recent progress, properties, applications, and future directions, Composites Part A: Applied Science and Manufacturing, Volume 165, 2023, 107362

[4] Rui Wang, Won Young Jang, Wen Zhang, Ch. Venkata Reddy, Raghava Reddy Kakarla, Changping Li, Vijai Kumar Gupta, Jaesool Shim, Tejraj M. Aminabhavi, Emerging two-dimensional (2D) MXene-based nanostructured materials: Synthesis strategies, properties, and applications as efficient pseudo-supercapacitors, Chemical Engineering Journal, Volume 472, 2023, 144913, ISSN 1385-8947

[5] Mawethu Pascoe Bilibana, Electrochemical properties of MXenes and applications, Advanced Sensor and Energy Materials, Volume 2, Issue 4, 2023

[6] Zhou J, Dahlqvist M, Björk J, Rosen J. Atomic Scale Design of MXenes and Their Parent Materials─From Theoretical and Experimental Perspectives. Chem Rev. 2023 Dec 13;123(23):13291-13322. doi: 10.1021/acs.chemrev.3c00241. Epub 2023 Nov 17. PMID: 37976459; PMCID: PMC10722466.

[7] Shoupik Mullani, Chihoon Kim, Vaibhav Lokhande, Taeksoo Ji, MXene structural and surface modifications for enhanced Li-ion diffusion in lithium-ion capacitors: A critical of recrepentanceChemical Engineering Journal, Volume 510, 2025

[8] Avishek Dey, Silvia Varagnolo, Nicholas P Power, Naresh Vangapally, Yuval Elias, Lois Damptey, Bright N. Jaato, Saianand Gopalan, Zahra Golrokhi, Prashant Sonar, Vimalnath Selvaraj, Doron Aurbach, Satheesh Krishnamurthy, Doped MXenes—A new paradigm in 2D systems: Synthesis, properties and applications, Progress in Materials Science, Volume 139, 2023

[9] Yuanlong Cui, Jie Zhu, Hui Tong, Ran Zou, Advanced perspectives on MXene composite nanomaterials: Types, synthetic methods, thermal energy utilization, and 3D-printed techniques, iScience, Volume 26, Issue 1, 2023

[10] H. Alnoor, A. Elsukova, J. Palisaitis, I. Persson, E.N. Tseng, J. Lu, L. Hultman, P.O.Å. Persson, Exploring MXenes and their MAX phase precursors by electron microscopy, Materials Today Advances, Volume 9, 2021

[11] Frédéric Payet. Exfoliation of functionalized layered oxides: synthetic approaches and advanced characterizations: toward a non-electrostatic reassembly. Material chemistry. Université de Strasbourg, 2022.

[12] Kumar S, Bhushan P, Bhattacharya S. Fabrication of Nanostructures with Bottom-up Approach and Their Utility in Diagnostics, Therapeutics, and Others. Environmental, Chemical, and Medical Sensors. 2017 Nov 18:167–98. doi: 10.1007/978-981-10-7751-7_8. PMCID: PMC7122830.

[13] Md Injamamul Haque Protyai, Adib Bin Rashid, A comprehensive overview of recent progress in MXene-based polymer composites: Their fabrication processes, advanced applications, and prospects, Heliyon, Volume 10, Issue 17, 2024

[14] Raghvendra Kumar Mishra, Jayati Sarkar, Kartikey Verma, Iva Chianella, Saurav Goel, Hamed Yazdani Nezhad, Exploring thetrtransformativeand multifunctional potential of MXenes in 2D materials for next-generation technology, Open Ceramics, Volume 18, 2024

[15] Raheela Akhter, Shrikant S. Maktedar, MXenes: A comprehensive review of synthesis, properties, and progress in supercapacitor applications, Journal of Materiomics, Volume 9, Issue 6, 2023

[16] Guohao Li, Shuhan Lian, Jie Wang, Guanshun Xie, Nan Zhang, Xiuqiang Xie, Surface chemistry engineering and the applications of MXenes, Journal of Materiomics, Volume 9, Issue 6, 2023

[17] Ishika Rana, Vishnu Kumar Malakar, Kumar Rakesh Ranjan, Chandrabhan Verma, Akram AlFantazi, Prashant Singh, Kamlesh Kumari, MXenes and their composites for high-performance detection of pharmaceuticals and pesticides: A comprehensive review, Composites Part B: Engineering, Volume 302, 2025

[18] Hafiz Taimoor Ahmed Awan, Muhammad Amirul Aizat Mohd Abdah, Maida Mehar, Rashmi Walvekar, Vishal Chaudhary, Mohammad Khalid, Ajit Khosla, MXene-polymer hybrid composites for advanced energy storage: Insights into supercapacitors and batteries, Journal of Energy Storage, Volume 95, 2024

[19] Liu, Yuhua & Zhang, Wei & Zheng, Weitao. (2022). Quantum Dots Compete at the Acme of MXene Family for the Optimal Catalysis. Nano-Micro Letters. 14. 10.1007/s40820-022-00908-3.

[20] Kvashina, T.S. & Uvarov, N. & Korchagin, M.A. & Krutskiy, Yu.L. & Ukhina, Arina. (2020). Synthesis of MXene Ti3C2 by selective etching of MAX-phase Ti3AlC2. Materials Today: Proceedings. 31. 592-594. 10.1016/j.matpr.2020.07.107.

[21] Rizwan, K., Khan, M. I., & Sajjad, M. T. (2023). *Handbook of Functionalized Nanostructured MXenes: Synthesis, Characterization, and Applications*. Springer Nature. ISBN: 978-3-031-27410-5.

[22] Keshav Narayan Alagarsamy, Leena Regi Saleth, Saravanan Sekaran, Laura Fusco, Lucia Gemma Delogu, Maksym Pogorielov, Açelya Yilmazer, Sanjiv Dhingra, MXenes as emerging materials to repair electroactive tissues and organs, Bioactive Materials, Volume 48, 2025

[23] Tijin Thomas, Sravani Bontha, Ankita Bishnoi, Pratibha Sharma, MXene as a hydrogen storage material? A review from fundamentals to practical applications, Journal of Energy Storage, Volume 88, 2024

[24] Shuck, Christopher & Ventura-Martinez, Kimberly & Goad, Adam & Uzun, Simge & Shekhirev, Mikhail & Gogotsi, Yury. (2021). Safe Synthesis of MAX and MXene: Guidelines to Reduce Risk During Synthesis. Journal of Chemical Health and Safety. 28. 326–338. 10.1021/acs.chas.1c00051.

[25] Kruger, Dawid & Garcia, Hermenegildo & Primo, Ana. (2024). Molten Salt Derived MXenes: Synthesis and Applications. Advanced Science. 11. 10.1002/advs.202307106.

[26] Xiu, Lu-Yang & Wang, Zhi-Yu & Qiu, Jie-Shan. (2020). General synthesis of MXene by green etching chemistry of fluoride-free Lewis acidic melts. Rare Metals. 39. 10.1007/s12598-020-01488-0.

[27] Mozafari, Mohammad & Soroush, Masoud. (2021). Surface functionalization of MXenes. Materials Advances. 2. 10.1039/D1MA00625H.

[28] Bashir, T., Ismail, S. A., Wang, J., Zhu, W., Zhao, J., & Gao, L. (2022). MXene terminating groups O, –F or –OH, –F or O, –OH, –F, or O, –OH, –Cl? *Journal of Energy Chemistry*, *76*, 90-104. https://doi.org/10.1016/j.jechem.2022.08.032

[29] Sahoo, S., Kumar, R., Hussain, I., & Zhang, K. (2024). Heteroatom doping in 2D MXenes for energy storage/conversion applications. *Advanced Powder Materials*, *3*(6), 100246. https://doi.org/10.1016/j.apmate.2024.100246

[30] Liu, Xinlong & Xu, Feng & Li, Zhiyong & Liu, Zhiting & Yang, Wei & Zhang, Yufei & Fan, Haosen & Yang, Hui. (2022). Design strategy for MXene and metal chalcogenides/oxides hybrids for supercapacitors, secondary batteries, and electro/photocatalysis. Coordination Chemistry Reviews. 464. 214544. 10.1016/j.ccr.2022.214544.

[31] Iravani, S., Zarepour, A., Nazarzadeh Zare, E., Makvandi, P., Khosravi, A., & Zarrabi, A. (2024). Synergistic advancements: Exploring MXene/graphene oxide and MXene/reduced

graphene oxide composites for next-generation applications. *FlatChem*, *48*, 100759. https://doi.org/10.1016/j.flatc.2024.100759

[32] Jiang, Y. (2025). Applications and perspectives of Ti3C2Tx MXene in electrochemical energy storage systems. *International Journal of Electrochemical Science*, *20*(2), 100948. https://doi.org/10.1016/j.ijoes.2025.100948

[33] Badawi, N., Bhuyan, M., Luqman, M., Alshareef, R. S., Rafe Hatshan, M., Al-Warthan, A., & Farooq Adil, S. (2024). MXenes: The future of solid-state supercapacitors: Status, challenges, prospects, and applications. *Arabian Journal of Chemistry*, *17*(9), 105866. https://doi.org/10.1016/j.arabjc.2024.105866

[34] Sohan, Arya & Banoth, Pravallika & Aleksandrova, Mariya & Grace, Andrews & Kollu, Pratap. (2021). Review on MXene synthesis, properties, and recent research exploring electrode architecture for supercapacitor applications. International Journal of Energy Research. 45. 10.1002/er.7068.

[35] Jangra, S., Kumar, B., Sharma, J., Sengupta, S., Das, S., Brajpuriya, R., Ohlan, A., Mishra, Y. K., & Goyat, M. (2024). A review on overcoming challenges and pioneering advances: MXene-based materials for energy storage applications. *Journal of Energy Storage*, *101*, 113810. https://doi.org/10.1016/j.est.2024.113810

[36] XU, HUIZHONG & Zheng, Dehua & Liu, Faqian & Li, Wei & Lin, Jianjian. (2020). Synthesis of MXene/Polyaniline Composite with Excellent Electrochemical Properties. Journal of Materials Chemistry A. 8. 10.1039/D0TA00572J.

[37] Cai, Yong-Zhu & Fang, Yong-Sheng & Cao, Wen-Qiang & He, Peng & Cao, Mao-Sheng. (2021). MXene-CNT/PANI ternary material with excellent supercapacitive performance driven by synergy. Journal of Alloys and Compounds. 868. 159159. 10.1016/j.jallcom.2021.159159.

[38] Wei, Yudi & Zheng, Mingming & Luo, Wenlong & Dai, Bo & Ren, Jiajia & Ma, Mingliang & Li, Tingxi & Ma, Yong. (2022). All pseudocapacitive MXene-MnO2 flexible asymmetric supercapacitor. Journal of Energy Storage. 45. 103715. 10.1016/j.est.2021.103715.

[39] Liu, S.; Zhang, H.; Chen, J.; Peng, X.; Chai, Y.; Shao, X.; He, Y.; Wang, X.; Ding, B. Functionalization Strategies of MXene Architectures for Electrochemical Energy Storage Applications. *Energies* **2025**, *18*, 1223. https://doi.org/10.3390/en18051223

[40] Ming, Fangwang & Liang, Hanfeng & Huang, Gang & Bayhan, Zahra & Alshareef, Husam. (2020). MXenes for Rechargeable Batteries Beyond the Lithium-Ion. Advanced Materials. 33. 10.1002/adma.202004039.

[41] Chy, M.N.U.; Rahman, M.A.; Kim, J.-H.; Barua, N.; Dujana, W.A. MXene as Promising Anode Material for High-Performance Lithium-Ion Batteries: A Comprehensive Review. *Nanomaterials* **2024**, *14*, 616. https://doi.org/10.3390/nano14070616

[42] Samejo, B. A., Naseer, K., Samejo, S., Janjhi, F. A., Memon, N., Castro-Muñoz, R., & Boczkaj, G. (2024). MXene-based composites for capacitive deionization – The advantages, progress, and their role in desalination - A review. *Water Resources and Industry*, *31*, 100230. https://doi.org/10.1016/j.wri.2023.100230

[43] Zhao, L., Zhang, T., Zhao, H., & Hou, Y. (2020). Polyanion-type electrode materials for advanced sodium-ion batteries. *Materials Today Nano*, *10*, 100072. https://doi.org/10.1016/j.mtnano.2020.100072

[44] Amani, A. M., Abbasi, M., Najdian, A., Mohamadpour, F., Kasaee, S. R., Kamyab, H., Chelliapan, S., Shafiee, M., Tayebi, L., Vaez, A., Najafian, A., Vafa, E., & Mosleh-Shirazi, S. (2025). MXene-based materials for enhanced water quality: Advances in remediation strategies. *Ecotoxicology and Environmental Safety*, *291*, 117817. https://doi.org/10.1016/j.ecoenv.2025.117817

[45] Zhang, P., Wang, Z., He, Z., Zhang, D., Wang, J., & Zhao, Y. (20The rese20 The research each of transition metal sulfide-based composite materials in sodium-ion batteries anodes: Heterostructure and theoretical calculation. *Composites Part B: Engineering*, *303*, 112564. https://doi.org/10.1016/j.compositesb.2025.112564

[46] Guo, Xin & Zhang, Jinqiang & Song, Jianjun & Wu, Wenjian & Wang, Guoxiu. (2018). MXene encapsulated titanium oxide nanospheres for ultra-stable and fast sodium storage. Energy Storage Materials. 14. 306-313. 10.1016/j.ensm.2018.05.010.

[47] Liu, Shude & Zhang, Huilin & Chen, Jieming & Peng, Xue & Chai, Yafei & Shao, Xian & He, Yi & Wang, Xiaoqiang & Ding, Bin. (2025). Functionalization Strategies of MXene Architectures for Electrochemical Energy Storage Applications. Energies. 18. 1223. 10.3390/en18051223.

[48] Sayed K, Aref M, G. Abokhalil A, A. Mossa M. Hybrid and Advanced Energy Storage Systems: Integration, Applications, and Future Trends [Internet]. Energy Storage Devices - A Comprehensive Overview [Working Title]. IntechOpen; 2025.

[49] Rudra, S., Seo, H. W., Sarker, S., & Kim, D. M. (2023). Supercapatteries as Hybrid Electrochemical Energy Storage Devices: Current Status and Future Prospects. *Molecules*, *29*(1), 243. https://doi.org/10.3390/molecules29010243

[50] K., Pavithra & Jeong, Sang Mun & Rout, Chandra. (2024). MXene-Carbon-based hybrid materials for supercapacitor applications. Energy Advances. 3. 10.1039/D3YA00502J.

[51] Eswaran, S. G., Rashad, M., Krishna Kumar, A. S., & M. EL-Mahdy, A. F. (2025). A Comprehensive Review of Mxene-Based Emerging Materials for Energy Storage Applications and Future Perspectives. *Chemistry – An Asian Journal*, *20*(4), e202401181. https://doi.org/10.1002/asia.202401181

[52] Varghese, S. M., Mohan, V. V., Suresh, S., Bhoje Gowd, E., & Rakhi, R. (2024). Synergistically modified Ti3C2Tx MXene conducting polymer nanocomposites as efficient electrode materials for supercapacitors. *Journal of Alloys and Compounds*, *973*, 172923. https://doi.org/10.1016/j.jallcom.2023.172923

[53] Balach, Juan & Giebeler, Lars. (2021). MXenes and the progress of Li–S battery development – a perspective. Journal of Physics Energy. 3. 021002. 10.1088/2515-7655/abd5c4.

[54] Zhu, Junjie & Avireddy, Hemesh & Biendicho, Jordi & Martinez-Soria, Luis & Rueda-Garcia, Daniel & Morante, J. & Ballesteros, Belen & Gómez-Romero, Pedro. (2022). Rational Design of MXene/Activated Carbon/Polyoxometalate Triple Hybrid Electrodes with Enhanced Capacitance for Organic-Electrolyte Supercapacitors. Journal of Colloid and Interface Science. 623. 10.1016/j.jcis.2022.04.170.

[55] Awan, Hafiz & Abdah, Muhammad & Mehar, Maida & W, Rashmi & Chaudhary, Vishal & Khalid, Mohammad & Khosla, Ajit. (2024). MXene-polymer hybrid composites for advanced energy storage: Insights into supercapacitors and batteries. Journal of Energy Storage. 95. 112449. 10.1016/j.est.2024.112449.

[56] Jangra, S., Kumar, B., Sharma, J., Sengupta, S., Das, S., Brajpuriya, R., Ohlan, A., Mishra, Y. K., & Goyat, M. (2024). A review on overcoming challenges and pioneering advances: MXene-based materials for energy storage applications. *Journal of Energy Storage*, *101*, 113810. https://doi.org/10.1016/j.est.2024.113810

[57] Behzadi Pour, G., Nazarpour Fard, H., & Fekri Aval, L. (2024). A Comparison of the Electrical Properties of Gel Polymer Electrolyte-Based Supercapacitors: A Review of Advances in Electrolyte Materials. *Gels*, *10*(12), 803. https://doi.org/10.3390/gels10120803

[58] Hussain, I., Arifeen, W. U., Khan, S. A., Aftab, S., Javed, M. S., Hussain, S., Ahmad, M., Chen, X., Zhao, J., Rosaiah, P., Fawy, K. F., Younis, A., Sahoo, S., & Zhang, K. (2024). M4X3 MXenes: Application in Energy Storage Devices. *Nano-Micro Letters*, *16*, 215. https://doi.org/10.1007/s40820-024-01418-0

[59] R.R. G, Kadathamana A, Varghese J, Lugo Loredo S, Cerdán-Pasarán A, Pushpan S, et al. Electrochemical Performance and Design Strategies of MXene-Based Supercapacitors [Internet]. Energy Storage Devices - A Comprehensive Overview [Working Title]. IntechOpen; 2024.

[60] Das, Pratteek & Wu, Zhong-Shuai. (2020). MXene for energy storage: Present status and future perspectives. Journal of Physics: Energy. 2. 10.1088/2515-7655/ab9b1d.

[61] Oyehan, Tajudeen & Salami, Babatunde & Abdulrasheed, Abdulrahman & Hambali, Hambali & Gbadamosi, Afeez & Valsami-Jones, Eugenia & Saleh, Tawfik. (2023). MXenes: Synthesis, properties, and applications for sustainable energy and environment. Applied Materials Today. 35. 101993. 10.1016/j.apmt.2023.101993.

[62] Jussambayev, M., Shakenov, K., Sultakhan, S., Zhantikeyev, U., Askaruly, K., Toshtay, K., & Azat, S. (2025). MXenes for sustainable energy: A comprehensive review on conservation and storage applications. *Carbon Trends*, *19*, 100471. https://doi.org/10.1016/j.cartre.2025.100471

[63] A. Fattah-alhosseini, Z. Sangarimotlagh, M. Karbasi, M. Kaseem, Review of MXene/MOF composites as photocatalysts for pollutant degradation, *Nano-Structures & Nano-Objects*, Volume 38, 2024, 101192, ISSN 2352-507X.

[64] J. Sengupta, C. M. Hussain, MXene-based electrochemical biosensors: Advancing detection strategies for biosensing (2020–2024), *Biosensors*, Volume 15, Issue 3, 2025, 127, ISSN 2079-6374.

[65] M. Tahir, U. Fatima, Recent trends and emerging challenges in two-dimensional materials for energy harvesting and storage applications, *Energy Storage*, Volume 4, 2021, e244, ISSN 2708-058X.

[66] Y. Guo, Z. Du, Z. Cao, B. Li, S. Yang, MXene derivatives for energy storage and conversions, *Small Methods*, Volume 7, Issue 6, 2023, 2201559, ISSN 2366-9608.

[67] R. M. Ronchi, J. T. Arantes, S. F. Santos, Synthesis, structure, properties and applications of MXenes: Current status and perspectives, *Ceramics International*, Volume 45, Issue 15, 2019, Pages 18167–18188, ISSN 0272-8842.

[68] X. Wang, F. Tang, Q. Cao, X. Qi, M. Pearson, M. Li, H. Pan, Z. Zhang, Z. Lin, Comparative study of three carbon additives: Carbon nanotubes, graphene, and fullerene-C60, for synthesizing enhanced polymer nanocomposites, *Nanomaterials*, Volume 10, Issue 5, 2020, 838, ISSN 2079-4991.